

WORK SHEET - 06

STRAIGHT LINE AND CIRCLE

[HINT SOLUTION]

Q.1 $\frac{7}{2}$

Sol. Let P $(2\cos \theta, 2\sin \theta)$ and Q $(-\cos \theta, -2\sin \theta)$ where $\theta \in [0, 2\pi)$

$$\text{So, } L_1 = \frac{|2\cos \theta + 2\sin \theta - 1|}{\sqrt{2}}, L_2 = \frac{|-2\cos \theta - 2\sin \theta - 1|}{\sqrt{2}}$$

$$\therefore L_1 L_2 = \frac{|(2\cos \theta + 2\sin \theta)^2 - 1|}{2} = \frac{|3 + 4\sin 2\theta|}{2} \leq \frac{7}{2}$$

So, maximum value of $L_1 L_2$ equals $\frac{7}{2}$.

Q.2 4

Sol. Clearly, C (x, y) lies on the circle described on AB as diameter.

Also, radius of circle = $\sqrt{2}$

Now, area $(\Delta ABC) = 1$

$$\Rightarrow \frac{1}{2} (2\sqrt{2}) (\text{height}) = 1$$

$$\Rightarrow \text{height} = \frac{1}{\sqrt{2}} < \sqrt{2} = \text{radius of circle}$$

So, by symmetry there will be 4 positions of C in xy plane.

Note: If area $(\Delta ABC) = 2 \Rightarrow$ altitude = radius.

So, two possible positions of C.

Aliter:

$$\Delta = 1 \quad (\text{Given})$$

$$\Rightarrow \begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 4 & 1 \end{vmatrix} = \pm 2 \Rightarrow x - y = 0 \text{ or } x - y = -2.$$

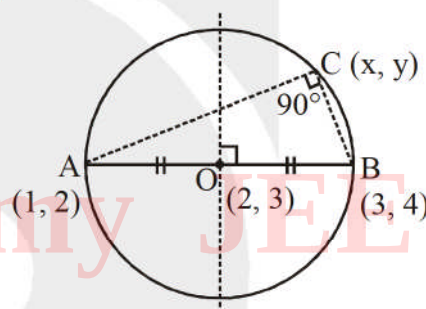
$$\text{Put } y = x \text{ in } (x - 1)(x - 3) + (y - 2)(y - 4) = 0.$$

We get two points C(x, y).

$$\text{Similarly, put } y = x + 2 \text{ in } (x - 1)(x - 3) + (y - 2)(y - 4) = 0.$$

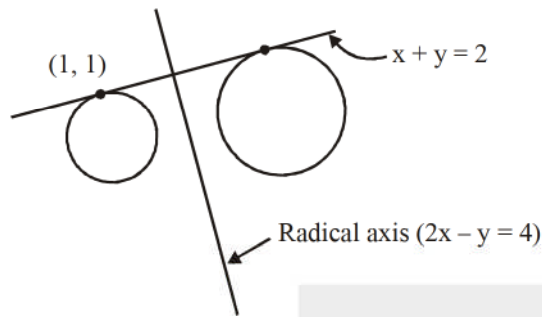
We get two more points C(x, y).

Hence, total 4 points C(x, y) are possible.



Q.3 $(3, -1)$ (C) $\left(\frac{5}{2}, \frac{-1}{2}\right)$ (D) $(2, 0)$

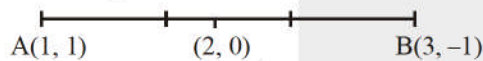
Sol. Radical axis always bisects the common tangents of two circles



∴ Point of intersection of common tangent and radical axis

$$\begin{cases} 2x - y = 4 \\ x + y = 2 \end{cases} \Rightarrow x = 2, y = 0$$

∴ Another point of contact is



∴ B ≡ (3, -1)

Q.4 1

Sol. Clearly, P remains in the shaded portion as shown in diagram.

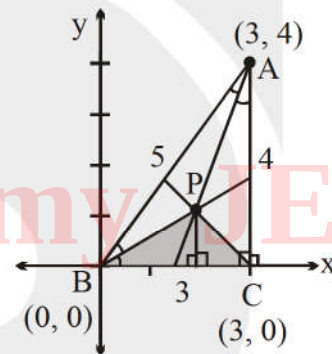
So, maximum value of d(P, BC)

= inradius

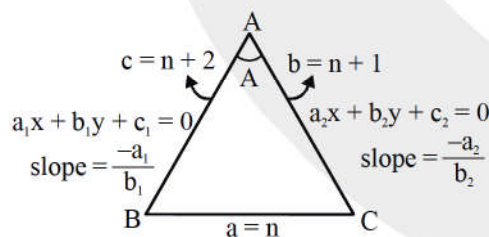
= radius of circle inscribed in ΔABC

$$\Rightarrow r = \frac{\Delta}{s} = \frac{\frac{1}{2}(3)(4)}{\left(\frac{3+4+5}{2}\right)} \quad [\text{Note: P is incentre of } \Delta ABC]$$

$$= \frac{6}{6} = 1.$$



Q.5 6



Sol.

$$\therefore \tan A = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\left(\frac{-a_1}{b_1} \right) - \left(\frac{-a_2}{b_2} \right)}{1 + \frac{a_1 a_2}{b_1 b_2}} \right| = \left| \frac{a_1 b_2 - a_2 b_1}{a_1 a_2 + b_1 b_2} \right|$$

$$\text{So, } \tan A = \frac{4}{3} \Rightarrow \cos A = \frac{3}{5}$$

$$\therefore \frac{(n+2)^2 + (n+1)^2 - n^2}{2(n+2)(n+1)} = \frac{3}{5} \quad (\text{using cosine law in } \triangle ABC)$$

$$\Rightarrow n = 13$$

$$\text{So, } \left(\frac{BC + CA - AB}{2} \right) = \left(\frac{13 + 14 - 15}{2} \right) = \frac{12}{2} = 6$$

Q.6 $\frac{1}{2\sqrt{2}}$

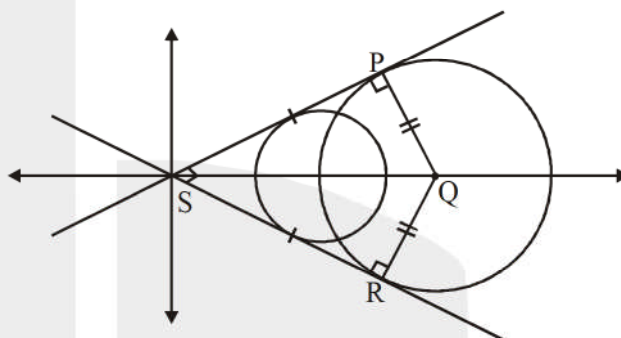
Sol. Clearly $\square PQRS$ is a square.

$$r = \sqrt{2} = SQ$$

$$\beta_n = \frac{1}{\sqrt{2}} \cdot \frac{1}{n(n+1)}$$

$$\beta_n = \frac{1}{\sqrt{2}} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$\sum_{n=2}^n \beta_n = \frac{1}{\sqrt{2}} \left(\frac{1}{2} - \frac{1}{n+1} \right) \Rightarrow \sum_{n=2}^{\infty} \beta_n = \frac{1}{2\sqrt{2}}$$



Q.7 4

Sol. $L: x - y = 0$

$$A \equiv (3, 1 + \sin \theta), B \equiv (-1, 1 + \cos \theta)$$

Both the points lying on opposite side of the line

$\therefore$ For least value of $(PA + PB)$ points A, P and B must be collinear

$$\Rightarrow AB = \sqrt{16 + (\sin \theta - \cos \theta)^2}$$

$$AB|_{\text{least}} = 4 \quad \text{when } \sin \theta - \cos \theta = 0 \Rightarrow \theta = \frac{\pi}{4}$$

Q.8 4

Sol. (a, b) must be foot of the perpendicular

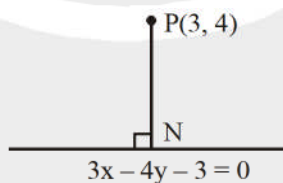
$$\frac{x-3}{3} = \frac{y-4}{-4} = - \left(\frac{9-16-3}{9+16} \right)$$

$$\frac{x-3}{3} = \frac{y-4}{-4} = \frac{2}{5}$$

$$\Rightarrow x = \frac{6}{5} + 3 = \frac{21}{5}$$

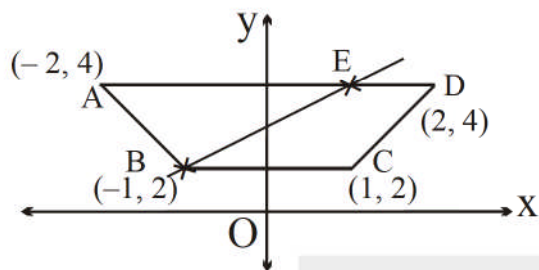
$$y = \frac{-8}{5} + 4 = \frac{12}{5}$$

$$\therefore a - b = \frac{21}{5} - \frac{12}{5} = \frac{9}{5} = \frac{p}{q} \Rightarrow p - q = 4$$



Q.9 $x - y + 3 = 0$

Sol.



The points A, B, C, D form a trapezium whose line of symmetry is the y-axis.
Let BE be the required line where $ED = \lambda$.

$$\text{Now, area (BCDE)} = \frac{1}{2} \text{ area (ABCD)} = \frac{1}{2} (2 + \lambda) 2 = \frac{1}{2} \left(\frac{1}{2} \right) (4 + 2) 2$$

$$\Rightarrow 2 + \lambda = 3 \Rightarrow \lambda = 1 = ED$$

$$\therefore E \equiv (1, 4)$$

$$\text{So, the required line BE is } (y - 2) = \frac{(4 - 2)}{1 - (-1)} (x + 1)$$

$$\Rightarrow y - 2 = x + 1 \Rightarrow x - y + 3 = 0.$$

Q.10 - 4

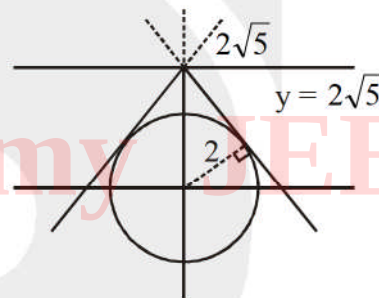
Sol. Tangent to the circle $x^2 + y^2 = 4$ from the point $(0, 2\sqrt{5})$

$$y - 2\sqrt{5} = m(x - 0)$$

$$mx - y + 2\sqrt{5} = 0$$

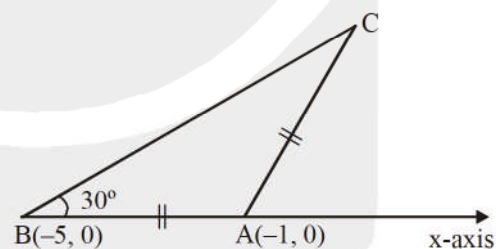
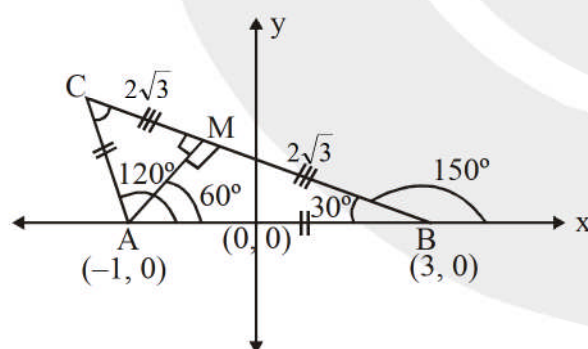
Applying, $p = r$

$$\left(\frac{2\sqrt{5}}{\sqrt{1+m^2}} \right) = 2 \Rightarrow 1 + m^2 = 5 \Rightarrow m^2 = 4 \Rightarrow m = \pm 2$$



Q.11 (A) $x + \sqrt{3}y = 3$ (C)

Sol.



$$\text{In } \triangle AMB, \frac{2\sqrt{3}}{AB} = \sin 60^\circ \Rightarrow \frac{2\sqrt{3}}{AB} = \frac{\sqrt{3}}{2}$$

$$\therefore AB = 4$$

$$\Rightarrow B(3, 0)$$

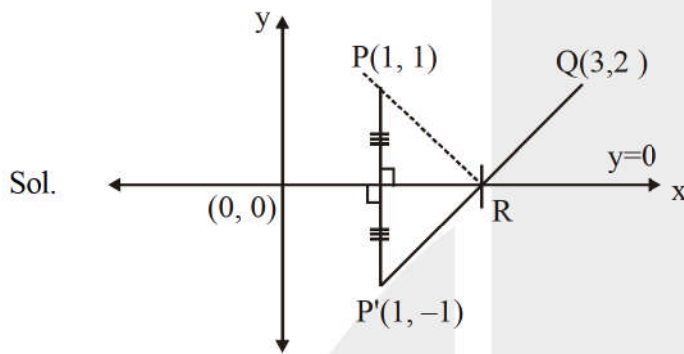
[Hint: Apply cosine rule in $\triangle ABC$
we get $AB = AC = 4$]

So, equation of BC is $(y - 0) = \frac{-1}{\sqrt{3}} (x - 3) \Rightarrow \sqrt{3}y = -x + 3$

$$\Rightarrow \boxed{x + \sqrt{3}y = 3}$$

and for second case $(y - 0) = \frac{1}{\sqrt{3}} (x + 5) \Rightarrow \boxed{x - \sqrt{3}y = 3} = 0.$

Q.12 (A)



The point R should be such that PR is reflected along RQ from the line $y = 0$.
The equation of P'Q (where P' is the image of P) is $3x - 2y = 5$

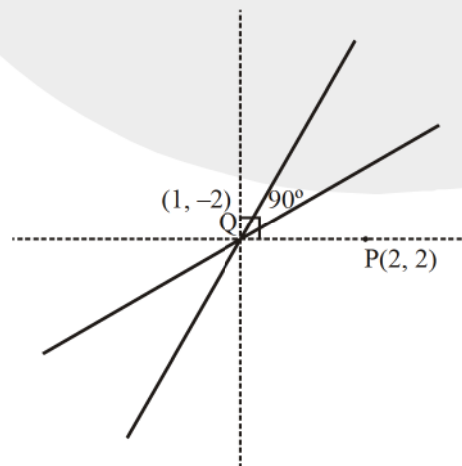
Clearly, $R \left(x = \frac{5}{3}, y = 0 \right)$ is the point of intersection of $y = 0$ and $3x - 2y = 5$.

Q.13 $x + 4y + 7 = 0$

Q.14 $\frac{49}{8}$

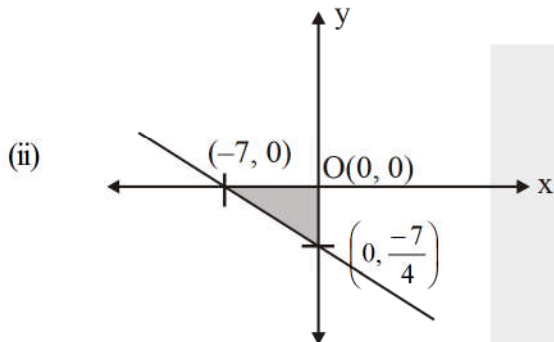
Q.15 5

Sol. Given, $(m + 2n)x + (m - 3n)y + (m - 8n) = 0, m, n \in \mathbb{R}$
 $\Rightarrow m(x + y + 1) + n(2x - 3y - 8) = 0$
 $\therefore$ Point of concurrency = $Q(1, -2)$



(i) $m_{PQ} = \frac{2+2}{2-1} = 4$

So, equation of line L is $(y+2) = \frac{-1}{4}(x-1)$
 $\Rightarrow 4y+8 = -x+1 \Rightarrow x+4y+7=0$. **Ans.**



So, $\Delta = \frac{1}{2} \left(\frac{7}{4} \right) (7) = \frac{49}{8}$. **Ans.**

(iii) The lines $x+4y+7=0$ and $x-2y+1=0$ intersect at $(-3, -1)$ which must satisfy the line $3x-4y+c=0 \Rightarrow -9+4+c=0 \Rightarrow c=5$. **Ans.**

Q.16 (C) 56 (D) 64

Sol. Equation of family of circle touching the line $x-y=0$ at $P(-4, -4)$ is

$$(x+4)^2 + (y+4)^2 + \lambda(x-y) = 0$$

$$x^2 + y^2 + (8+\lambda)x + (8+\lambda)y + 32 = 0.$$

$$r = \sqrt{\left(\frac{8+\lambda}{2}\right)^2 + \left(\frac{8-\lambda}{2}\right)^2} - 32$$

$$p = \left| \frac{-8-\lambda}{2} - \frac{8+\lambda}{2} \right| \frac{1}{\sqrt{2}} = \frac{16}{2\sqrt{2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$$

$$r^2 = p^2 + (3\sqrt{2})^2$$

$$\frac{64+\lambda^2+64+\lambda^2-128}{4} = 32+18$$

$$\frac{2\lambda}{4} = 50$$

$$\lambda^2 = 100 \Rightarrow \lambda = \pm 10.$$

Hence, 2 circles are possible

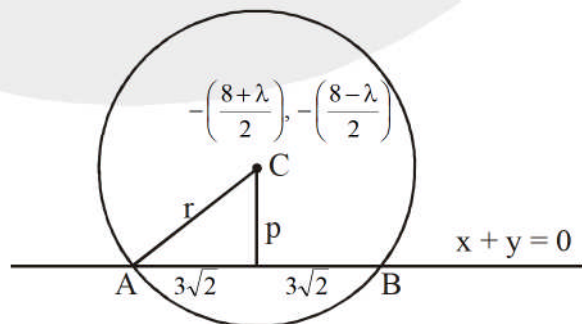
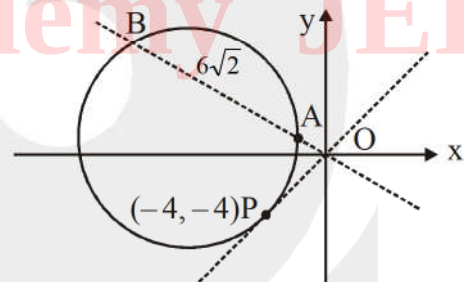
$$S_1 : x^2 + y^2 + 18x - 2y + 32 = 0$$

$$\text{and } S_2 : x^2 + y^2 - 2x + 18y + 32 = 0$$

but $(-10, 2)$ lies inside $S_1 = 0$.

Hence, $a = -2$, $b = 18$, $c = 32$

$$\therefore (a+b+c) = 48.$$



Q.17 (A) $x - 3y + 1 = 0$ (D) $3x + y - 12 = 0$

Sol.

The equation of side BC will be perpendicular to the bisectors of $\angle BAC$.

Now, the equations of the bisectors of lines AB and AC are $\frac{(x+y-5)}{\sqrt{2}} = \pm \frac{(7x-y-3)}{5\sqrt{2}}$

$$\Rightarrow x - 3y + 11 = 0 \text{ or } 3x + y - 7 = 0$$

Let the equation of side BC be $x - 3y + \lambda = 0$ and altitude through vertex A be AD.

Then equation of AD is $3x + y - 7 = 0$. If $AD = l$, then

$$\text{area } (\triangle ABC) = 5 \Rightarrow \frac{1}{2} (BC) (AD) = 5$$

$$\Rightarrow \frac{1}{2} (2l |\tan \theta|) (l) = 5 \Rightarrow l^2 |\tan \theta| = 5$$

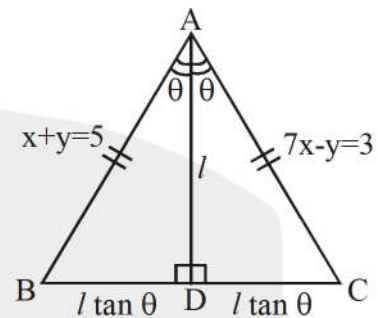
$$\Rightarrow l^2 \left(\frac{1}{2} \right) = 5 \quad \left(\text{As, } |\tan \theta| = \left| \frac{-3+1}{1+3} \right| = \frac{1}{2} \right)$$

$$\Rightarrow l^2 = 10 \Rightarrow \left(\frac{|11-\lambda|}{\sqrt{10}} \right)^2 = 10$$

$$\Rightarrow |11 - \lambda|^2 = 100 \Rightarrow \lambda = 1, 21$$

$\therefore$ The equation of BC is $x - 3y + 1 = 0$ or $x - 3y + 21 = 0$.

Similarly, if equation of BC is $3x + y + \lambda = 0$, then equation of AD will be $x - 3y + 11 = 0$.



Now, $|\tan \theta| = \left| \frac{7-\frac{1}{3}}{1+\frac{3}{7}} \right| = 2$

$$\text{Now, area } (\triangle ABC) = 5 \Rightarrow \frac{1}{2} (BC) (AD) = 5$$

$$\Rightarrow \frac{1}{2} (2l |\tan \theta|) (l) = 5 \Rightarrow l^2 |\tan \theta| = 5$$

$$\Rightarrow 2l^2 = 5 \Rightarrow l^2 = \frac{5}{2} \Rightarrow \left(\frac{|3+4+\lambda|}{\sqrt{10}} \right) = \frac{5}{2}$$

$$\Rightarrow (\lambda + 7)^2 = 25 \Rightarrow (\lambda + 7) = \pm 2$$

$$\therefore \lambda = -2, -12$$

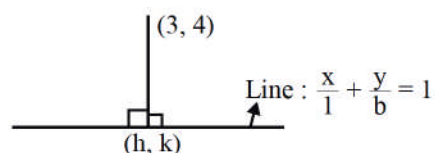
$\therefore$ The equation of BC is $3x + y - 2 = 0$ or $3x + y - 12 = 0$.

Hence, there will be 4 possible equation of side BC.

Q.18 (B) Radius of circle $\sqrt{5}$.

(C) If tangent is drawn to $P(x, y) = 0$ from the origin, then the length of tangent is $\sqrt{3}$.

Sol.



Now, $\left(\frac{k-4}{h-3}\right) \cdot \left(\frac{-1}{1/b}\right) = -1 \Rightarrow b = \left(\frac{h-3}{k-4}\right) \dots\dots(i)$

Also, $h + \frac{k}{b} = 1 \dots\dots(ii)$

$\therefore$ From (i) and (ii), we get locus as

$$(x-2)^2 + (y-2)^2 = (\sqrt{5})^2$$

Q.19 4

Sol. Equation of S_3 is $(x-0)(x-1) + (y-0)^2 + \lambda y = 0$
i.e. $x^2 + y^2 - x + \lambda y = 0$

i.e. $\sqrt{\frac{1}{4} + \frac{\lambda^2}{4}} - 0 = 1 \Rightarrow \lambda = \pm\sqrt{3}$

$\therefore$ Circle S_3 lie above x-axis $\Rightarrow \lambda = -\sqrt{3}$

i.e. $S_2 \equiv x^2 + y^2 - x - \sqrt{3}y = 0$

its centre $S_3 \equiv \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

Slope of line joining c_1 & $c_2 = \sqrt{3}$

$\therefore$ Slope of common tangent is $\sqrt{3}$

i.e. $\sqrt{3}x - y + k = 0$ is required common tangent and touches circle S_1

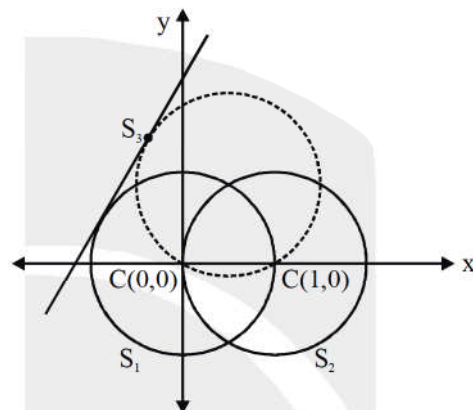
$\Rightarrow \left| \frac{k}{\sqrt{3+1}} \right| = 1 \Rightarrow k = \pm 2$

$\therefore$ For the given figure $k = 2$

$\therefore$ Required common tangent is $\sqrt{3}x - y + 2 = 0$

$\therefore a = \sqrt{3}, b = -1,$

$\therefore (a^2 - b) = 3 + 1 = 4$ **Ans.**



Q.20 17

Sol. Given, circles are

$x^2 + y^2 + 2gx + 2fy + c = 0 \dots\dots(1)$

$x^2 + y^2 = 4$; centre $(0, 0) \dots\dots(2)$

$x^2 + y^2 - 6x - 8y + 10 = 0$, centre $(3, 4) \dots\dots(3)$

$x^2 + y^2 + 2x - 4y - 2 = 0$; centre $(-1, 2) \dots\dots(4)$

Now, radical axis of (1) and (2) is $4 + 2gx + 2fy + c = 0$

As, $(0, 0)$ satisfy it $\Rightarrow c = -4 \dots\dots(5)$

Also, radical axis of (1) and (3) is

$(2g + 6)x + (2f + 8)y + (c - 10) = 0$

As, $(3, 4)$ satisfy it and using $c = -4$, we get

$3g + 4f = -18 \dots\dots(6)$

and radical axis of (1) and (4) is $(2g - 2)x + (2f + 4)y + (c + 2) = 0$

As, $(-1, 2)$ satisfy it and using $c = -4$, we get

$$2g - 4f = 8$$

.....(7)

$\therefore$ On solving (6) and (7), we get $g = -2, f = -3$

So $g^2 + f^2 - c = 4 + 9 - (-4) = 17$ **Ans.**

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